Efficient Few-Shot Gujarati to English Machine Translation using Quantized LLaMA Models and LoRA Fine-Tuning

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Abstract Machine translation serves an important purpose in facilitating communication between languages, especially those that are under-resourced like Gujarati, where data sets are scarce. Recent developments in the field of Natural Language Processing (NLP) include Large Language Models (LLMs) that perform sophisticated language functions with exceptional precision. In particular, transformer models like the Meta Llama 3.1 model have been found to perform satisfactorily in machine translation and multilingual comprehension tasks. Fine-tuning such large models may prove to be costly, requiring extensive computational resources and memory.The present project suggests a technique for efficient Gujarati to English machine translation based on few-shot learning with quantized LLaMA models and Low-Rank Adaptation (LoRA) fine-tuning. Quantized models have smaller memory footprints than their full-precision counterparts and require fewer computational resources. LoRA permits efficient fine-tuning through partial updates of a small fraction of model parameters. Few-shot learning can train models with relatively low amounts of data. BLEU score is used to assess the performance of our system, which compares the machine-translated sentences to the ground truth sentences and assigns a score based on their similarity.The experimental evaluation demonstrates that the few-shot (3-shot) configuration significantly outperforms the zero-shot setting across multiple evaluation metrics. Notably, the BLEU score improved from 3.43 to 13.77, while the TER value decreased from 115.39 to 90.16, indicating enhanced translation accuracy and reduced edit distance. Additionally, improvements in chrF, METEOR, and ROUGE-L scores confirm better semantic alignment and fluency in the generated translations, demonstrating the effectiveness of the proposed quantized LLaMA with LoRA fine-tuning approach for low-resource Gujarati-to-English machine translation.

Keywords: Machine Translation · Quantization · LoRA Fine-Tuning.

1 Introduction

Machine Translation (MT) has become an essential tool for modern communication where two different languages speaking individuals can communicate without any hassle. Though a lot of progress has been made in the field of Natural Language Processing (NLP), building good translation tools for low-resource languages such as Gujarati remains extremely challenging because of the lack of annotated data and need for more computational power. Advances in the field of Large Language Models (LLM)Tom Brown et al.[12], like transformer architecture in Meta Llama 3.1, can produce outstanding performance in terms of language translation.

However, such large models require significant computational resources.There are various ways to address this problem, among which are model quantization and Low-Rank Adaptation (LoRA) techniques Hu et al.[8] that significantly reduce memory consumption and enable fine-tuning of the model without modification of most of its parameters. Few-Shot Learning enables the model training process with only a few samples, which makes it possible to fine-tune the model on low-resource language data. In this project, an efficient MT system based on quantized LLaMA models[6][7] and LoRA fine-tuning is developed for Gujarati-English language pair and evaluated using BLEU metric.

Regarding our contribution to the project, we were tasked with developing and implementing the proposed method for translating from Gujarati to English based on a quantized version of the LLaMA architecture and parameter-efficient fine-tuning via LoRA. As a group, we cooperated in the preparation of the parallel corpora for the Gujarati-English language pair and setting up the few-shot learning setup. Furthermore, we all participated in conducting several experiments and analyzing the results obtained through different evaluation metrics, including BLEU, chrF, METEOR, ROUGE-L, and TER.

2 Literature Review

Machine Translation has evolved from being rule based to statistical and now to neural methods. The neural models have also improved in terms of context understanding and fluency in translation because of multilingual datasets[18] now available. However, even though translation of high-resource languages performs well, but low resource languages like Gujarati don't as there is not much data available in comparison with other resource-rich languages. The primary reasons include lack of parallel datasets available and linguistic complexity which results in inaccurate translation.

Transformer as introduced by Vaswani et al.[1] uses self-attention instead of RNN/CNN. This has many advantages such as capturing longer dependencies and faster training. It became the foundation for modern Machine Translation Systems and LLMs like LLaMA. However, it has a few major limitations, i.e., it needs huge amounts of data and compute.

To bring the problems, which came because of huge multilingual datasets, into use, different models were proposed. One was, mBART [2], trained on multiple languages and helps with the use of low-resource languages. Other was IndiTrans [3], designed for Indian languages and gave better results when using Gujarati. These showed that multilingual training can help in various ways.

Data needs to be readily available in huge amounts and so it is still an important factor in the performance of translation systems. Some multilingual and Indian languages datasets are available like AI4Bharat[4] etc [16][17]. These helped improving the performance of the systems. However, low-resource languages like Gujarati still have a problem that the data is limited for them. So, there is a need for better resources focusing on such low-resource languages.

Rare words posed a major challenge in neural machine translation. This is common in morphologically rich languages like Gujarati which also has a complex structure. This was addressed by Sennrich et al [5]. They proposed Byte Pair Encoding (BPE) to capture rare or unseen words more effectively and so handling morphology much better, beneficial and important for Gujarati.

With a massive jump in new LLMs, NLP tasks have transformed including machine translation. Models such as LLaMA, developed by Meta AI [6] [7], proved that smaller and efficiently trained models can give better results, achieving competitive performance. These models are available in various parameter sizes (3B or 8B). This is advantageous for research environments with limited resources. On the contrast, large-scale multilingual models such as BLOOM [11] and GPT3 [12], even though showing strong capabilities in cross-lingual understanding and generation, require huge compute and are not optimized for low-resource language pairs.

For improving efficiency, parameter-efficient fine-tuning methods [10] were developed. Low-Rank Adaptation (LoRA) as proposed by Hu et al [8], finetunes by only updating a small subset of parameters, hence reducing memory and compute requirements significantly while maintaining performance. Taking it further, Dettmers et al [9] introduced QLoRA, combining quantization with LoRA for quantized models. These methods are high beneficial for low compute environments. Also, quantization techniques like 8-bit optimizers[13], helps in enhancing efficiency as they reduce model weights without having significant impact on performance.

Machine Translation Systems are generally evaluated using standardized metrics such as BLEU score [14] using FLORES [15]. BLEU score measures similarity with reference translation. FLORES dataset is a benchmark for multilingual tasks. This helps in comparing different models without bias.

Even after so much enhancements and advancements, many limitations remain to be reseached. Most of the current methods either depend on large-scale datasets or require significant compute, making them less suitable for low-resource environments. Adding to that, limited work has been done to explore the use of parameter-efficient fine-tuning techniques, such as LoRA, combining with quantized large language models for Indic languages. To be more precise, the use of few-shot learning with quantized LLaMA models for Gujarati to English translation remains to be explored to a huge extent.

To address the above problems, this work focuses on developing an efficient Gujarati to English translation system using quantized LLaMA models with LoRA-based fine-tuning. The proposed method takes advantage of few-shot learning to handle lack of data problem while maintaining computational effi-

Table 1. Sample Gujarati-English sentence pairs from the dataset

Gujarati Sentence	English Translation
સેન્ટ માર્ક્સ રોડ	St Marks Road
આ બંનેની ઓળખાણ એક કોમન ફ્રેન્ડ દ્વારા થઇ હતી.	The two have supposedly met through a common friend.
તો આ સમાચાર સાંભળીને તમને પણ આશ્ચર્ય થશે!	Shocked to hear this news!
તમાકુ ફેફસાંને નુકસાન પહોંચાડે છે.	Tobacco damages the lungs.
વડાપ્રધાન નરેન્દ્ર મોદીએ મહત્વપૂર્ણ નિવેદન આપ્યું.	Prime Minister Narendra Modi made an important statement.

ciency, and hence adding value to the evolvement of machine translation for low-resource Indic languages.

3 Methodology

3.1 Dataset and Preprocessing

A dataset containing Gujarati-English parallel sentence pairs is used in this study. In it, each Gujarati sentence is aligned with its English translation. The origin of the dataset publicly sources and is a curated dataset [16] [17]. After this, preprocessing is done. It ensured better data quality, like clean text, normalization and, structured and formatted input-output pairs. A small subset of data is taken and formatted to guide the model during training and inference, for few-shot learning. In this study, an efficient framework for Gujarati to English translation using LLMs is proposed by us. The main objective is to create such a system that can work even in a low compute environment. A combination of quantized LLaMA models and LoRA-based fine-tuning methods is used, so that memeory requirements are minimum and efficiency can be improved. Along with this, a few-shot strategy is used to address the problem of scare availability of parallel Gujarati-English data. In a nutshell, the workflow includes data pre-processing, model setup, efficient fine-tuning, and evaluation using standard metrics.

3.2 Proposed Architecture

LLaMA [6] [7], built on transformer architecture, is used as the base architecture in this work. It is done because of its efficiency and strong performance in language understanding tasks. This model is also available in multiple parameter sizes like 3B or 8B, and is already pre-trained on large-scale text data, making it efficient in capturing linguistic patterns. The biggest advantage of using this in this study is that it satifies our main objective as it has comparatively lower compute requirements compared to other LLMs. The translation task can be formulated as:

$$\hat{y} = \arg\max_{y} P(y \mid x)$$

where x represents the Gujarati input sentence and y represents the generated English translation. The Transformer architecture computes contextual relationships using an attention mechanism defined as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

Quantization techniques [13] are used to mitigate the high computational requirements of LLMs. As a result, there is a significant decrease in the memeory usage and compute cost because quantization reduces the model-weight's precision, generally to a 4-bit or 8-bit representations. And so now, the model can be deployed and executed even on low compute hardwares. Even with reduced precision, the degradation in model performance is minimal. This makes it a practical method for this study.

$$W_q = Q(W)$$

where W represents the original weights and Wq represents the quantized weights. This enables the model to be deployed and executed even on low-compute hardware. Even with reduced precision, the degradation in model performance is minimal, making it a practical method for this study.

Now, LoRA-based fine-tuning [8] is used to improve the efficiency even more. Using this, there is a significant reduction in the number of trainable parameters, alongside maintaining performance.This is made possible because LoRA introduces trainable low-rank matrices into specific layers of the network, instead of updating all the parameters. In this way LoRA ensures efficient adaptation of LLMs for translation tasks, by freezing majority weights.

$$\Delta W = AB$$

$$W' = W + \Delta W$$

where A and B are low-rank matrices such that r d. This ensures efficient adaptation of large language models for translation tasks while freezing the majority of weights.

To address the problem of limited availability of Gujarati-English data, a few-shot approach [12]is used. It helps the model learn translation pattern and generalization capability.It is done by providing a small number of example translations to the model for context.

$$\mathcal{P}(x) = \{(x_1, y_1), (x_2, y_2), \ldots, (x_k, y_k), x\}$$

An efficient-parameter setup combining LoRA with the quantized LLaMA model is used for the training process. Learning rate, batch size and number of epochs are the key hyperparameters which are changed based on experimental requirements. Hence, implementation is done using PyTorch and Hugging face transformers. Such a balanced configuration is designed to enable efficient compute and model performance.

$$\mathcal{L}(\theta) = -\sum_{i=1}^{N} \log P(y_i \mid x_i; \theta)$$

BLEU score [14], a widely used metric in machine translation, is used to measure the performance of this proposed system. Measuring the similarity between machine-generated and reference translations based on a n-gram overlap, its higher value indicates better translation quality. This helps in standardized comparison with different models.

$$\text{BLEU} = BP \cdot \exp\left(\sum_{n=1}^{4} w_n \log p_n\right)$$

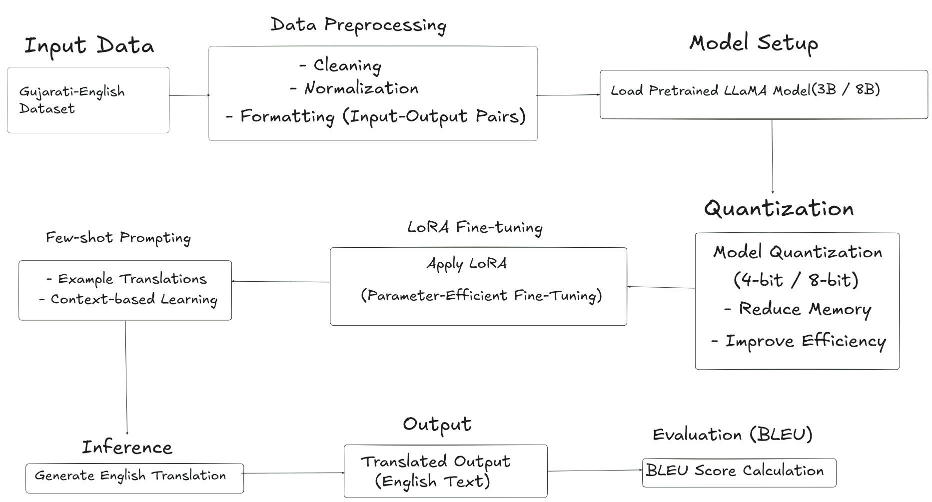

Figure 1. Architecture of the proposed Gujarati to English translation system

4 Results and Discussion

4.1 Quantitative Analysis

Multiple quantitative metrics have been used to evaluate the performance of the proposed Gujarati-English translation system. Experiments have been done

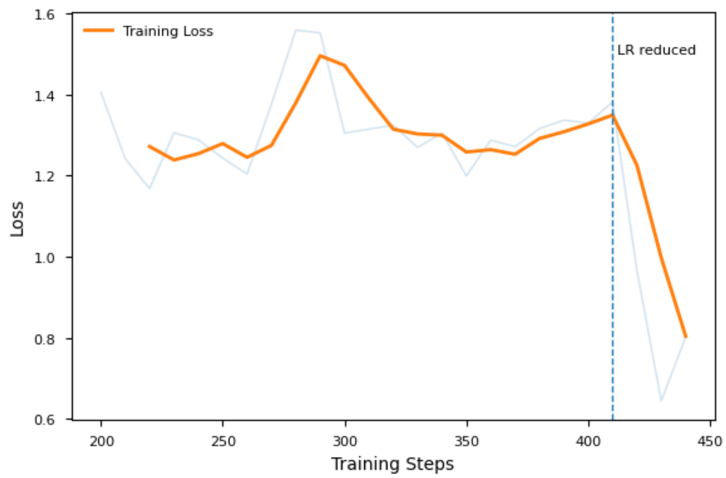

Figure 2. Training loss curve during fine-tuning

Table 2. Performance comparison between zero-shot and 3-shot settings

Metric	Zero-Shot	3-Shot
BLEU Score	3.43	13.77
chrF Score	8.53	39.29
METEOR Score	17.45	40.10
ROUGE-L	14.01	38.17
TER (↓)	115.39	90.16

under two separate settings: zero-shot (no examples are provided to the model) and few-shot (3-shot, where a small number of examples are provided). Few-shot approach has been done mostly because Gujarati is a low-resource language, and so this approach improves the performance of the system. Here, the numerical performance metrics are discussed to show the performance of the system.

A loss curve is used to analyze the training performance, showing model improvements over a period of time. As expected in the early stages of training, the training loss shows fluctuations, which can be seen in the Fig. 2. The model then starts learning meaningful pattern and so the loss gradually decreases. However, a significant reduction in loss was observed after adjusting the learning rate, proving advantages of training configuration. The terminal decrease in loss indicates convergence and better generalization capability of the model.

BLEU, chrF, METEOR, ROUGE-L, and TER are just a few of the many evaluation metrics used to measure translation performance. These metrics give a full picture of translation quality based on n-gram overlap, semantic similarity, and edit distance.

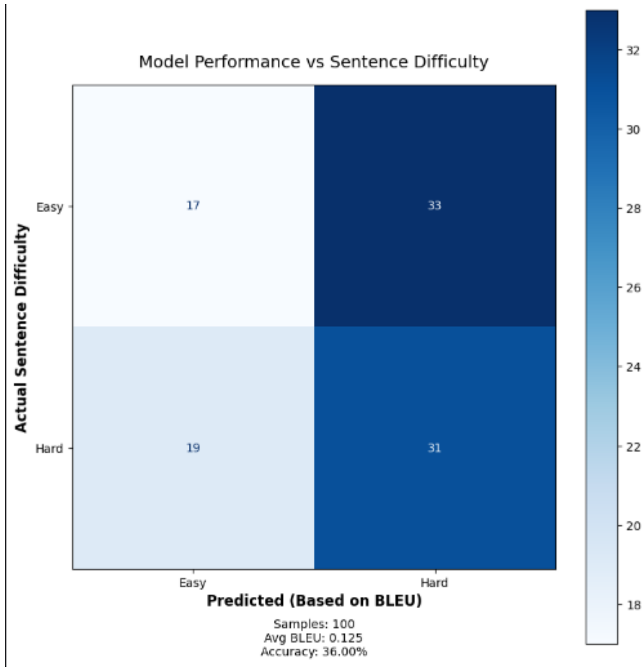

Figure 3. Confusion matrix based on sentence difficulty

The results show that the few-shot (3-shot) setting does much better than the zero-shot setting on all evaluation metrics. The BLEU score, for example, goes up from 3.43 to 13.77, which shows a big improvement in the quality of the translation. Likewise, higher chrF, METEOR, and ROUGE-L scores show that the translations are more semantically aligned and flow better.

The TER score also goes down from 115.39 to 90.16, which means that the edit distance is shorter and the translation is more accurate. This shows that adding a few examples helps the model learn how to translate better.

Taking the analysis of the model performance even further, a confusion matrix is created (Fig 3). It can be observed that the model performs better on simpler sentences compared to complex ones. This is due the linguistic complexity of complex sentences. This also highlights the limitation of the proposed model, having problem with complex grammatical structures and long sentences.

4.2 Qualitative Analysis of Translations

The model-generated translations and reference translations are used to perform a qualitative analysis. These results show that model can capture the meaningful translations of simpler sentences accurately. This is accompanied by negative effect in a way that the translations may contain grammatical errors or some partial information loss. These are in line with the limitations observed in the quantitative evaluation metrics.

5 Conclusion

The paper introduces an effective method of machine translation between Gujarati and English via quantized LLaMA models with LoRA-based fine-tuning. The proposed approach uses few-shot learning to overcome the issue of small amounts of data. The experimental evidence shows that the model can be used in low-resource settings with reasonable performance, which has been proven by the BLEU scores and qualitative analysis.

Quantization and parameter-efficient fine-tuning greatly decrease computational needs without compromising the quality of translation. Nonetheless, the model has weaknesses in its ability to deal with intricate sentence constructions, which means that it can be improved.

Future research can be aimed at refining the quality of translation based on larger datasets, more effective fine-tuning methods and the use of more sophisticated evaluation measures. On the whole, the work can be said to have contributed to the creation of effective machine translation systems in low resource languages.

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Table 3. Comparison of Gujarati input, reference translation, and model output

Gujarati Input	Reference Translation	Model Output
તેમને વેન્ટીલેટર પર રાખવામાં આવ્યા હતા.	They were kept on a ventilator.	He was kept on ventilator.
પેકેજ કે જે તમારી સિસ્ટમ પર બદલાયેલ છે તે શોધવામાં આવતું નથી.	The package that has been modified on your system cannot be found.	The package is not found on your system or in the repository.
વડાપ્રધાન નરેન્દ્ર મોદીએ દુનિયાના 10 સૌથી શક્તિશાળી નેતાઓમાં સ્થાન મેળવ્યું છે.	Prime Minister Narendra Modi has been ranked among the world's 10 most powerful leaders.	Prime Minister Narendra Modi is among the world's most powerful leaders.
આરએસએસના વિચારક અને ભાજપના રાજ્યસભાના સાંસદ રાકેશ સિંહે નિવેદન આપ્યું.	RSS ideologue and BJP Rajya Sabha MP Rakesh Singh gave a statement.	RSS ideologue and BJP Rajya Sabha MP Rakesh Singh said.
તમાકુ માત્ર ફેફસાંને નુકસાન નથી પહોંચાડતી તે હૃદયને પણ અસર કરે છે.	Tobacco not only damages the lungs but also affects the heart.	Smoking damages not only the lungs but also the heart.
હાલમાં, આ વીડિયો સોશિયલ મીડિયા પર પ્રશંસાનો વિષય બની ગયો છે.	The video is currently going viral on social media.	The video is currently trending on social media platforms.
આ કેસની સુનાવણી સુપ્રીમ કોર્ટની એક કોન્સ્ટિટ્યુશનલ બૅન્ચ કરી રહી છે.	The case is being heard in the constitutional bench of the apex court.	The case is being heard by the Supreme Court's constitutional bench.
ત્યાર બાદ શું થાય છે જે જાણવા તમારે ફિલ્મ જોવી પડશે.	What happens next you have to watch the film for that.	What happens next is what you need to watch the film for.
અરજીનાં એક વ્યક્તિએ પોતાની પત્ની પર ક્રુરતાનો આરોપ લગાવ્યો હતો.	The petitioner husband accused his wife of adultery.	One of the applicants accused his wife of cruelty.
રિઝર્વ બેન્ક ઓફ ઇન્ડિયાએ આ અંગેની માહિતી આપી.	This has been revealed by the Reserve Bank of India.	The Reserve Bank of India (RBI) said in its statement.
તેના લીધે તમને ચક્કર આવશે.	This will make you jittery.	It will make you feel dizzy.
ગુજરાત, મહારાષ્ટ્ર સહિતના રાજ્યોમાંથી મોટી સંખ્યામાં ઉત્તરપ્રદેશ, બિહાર, ઓડિશા વગેરે રાજ્યોના પ્રવાસી મજૂરો પોતાના વતન પરત ફર્યા છે.	Workers from Uttar Pradesh, Bihar, Odisha, Andhra Pradesh migrate mostly to Maharashtra and Gujarat, it noted.	A large number of migrant workers from Uttar Pradesh, Bihar, Odisha and other states have returned home in Gujarat, Maharashtra among others.

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